

Supplement: Forward Model Inputs

What counts as an input?	A value passed to the reported model’s forward call.
What is excluded?	The receiving model’s own targets and outputs; transferred parameters; cached arrays not passed forward; stopped candidates; upstream inputs of a frozen output.
Which widths are reported?	Active pre-projection channels in the fitted model contract.
What does a dash mean?	No value enters by that route.

Which reported model or form?	Which class implements it?	What object reaches the model?	What does it predict?
ShieldingTfnModel	ShieldingModel; RankRoutedPeptide ShieldingModel	One prepared structure; one of four input conditions	ORCA total shielding: one T_0 and five T_2 components
ShiftTfnModel (Static)	SingleConformationShiftModel	One statically prepared structure	Experimental shifts for nine atom roles
ShiftTfnModel (First retained frame)	SingleConformationShiftModel	First retained GROMACS frame, 150 ps	Experimental shifts for nine atom roles
ShiftMDModel	TrajectoryShiftModel	100 retained GROMACS frames; frozen first-frame parent response	Experimental shifts for nine atom roles

Table 1: Reported model names and implementations.

Which input condition?	What is added to Typed Graph?	Which ShieldingTfnModel input?	Which ShiftTfnModel comparison input?
Typed Graph	–	Shared structure only	Shared structure only
AIMNet2	256-value per-atom AIMNet2 representation	Shared structure + AIMNet2	Shared structure + AIMNet2
MOPAC	Selected PM7/MOZYME quantities	Shared structure + broad MOPAC surface	Shared structure + abbreviated MOPAC surface
AIMNet2+MOPAC	Both electronic additions	Shared structure + broad MOPAC surface + AIMNet2	Shared structure + abbreviated MOPAC surface + AIMNet2

Table 2: Reported input-condition names.

ShieldingTfnModel input conditions

Which condition?	What labels enter?	What are the node widths (scalar/flag/vector/ T_2)?	What are the MOPAC-pair widths (radial/scalar/vector/ T_2)?	What electronic addition enters?
Typed Graph	17	35 / 8 / 1 / 0	0 / 0 / 0 / 0	–
AIMNet2	17	291 / 8 / 1 / 0	0 / 0 / 0 / 0	256-value atom representation
MOPAC	17	96 / 12 / 23 / 18	11 / 2 / 3 / 1	Broad PM7/MOZYME node and pair surface
AIMNet2+MOPAC	17	352 / 12 / 23 / 18	11 / 2 / 3 / 1	Broad PM7/MOZYME surface + 256-value atom representation

Table 3: ShieldingTfnModel forward-input widths.

Which block?	What reaches the forward call?	Which conditions receive it?	Which direct method source applies?
Shared structure	The 17 labels, centred coordinates, covalent and residue relations, 35 scalars, eight applicability flags and one vector listed on the shared-structure page	All four	–
AIMNet2 atom representation	One saved 256-value internal atom representation; no other AIM-Net2 output	AIMNet2; AIMNet2+MOPAC	[3]
Broad MOPAC surface	The node and pair blocks on the next page	MOPAC; AIMNet2+MOPAC	MOPAC/PM7 [4, 5]
Graph geometry	Ten finite radial channels; seven bond-order indicators; nine bond-category indicators; same- and sequential-residue indicators; real spherical harmonics through rank 2	All four	–
Forced graph edges	Every covalent edge; additionally every reported MOPAC pair in MOPAC conditions	All four (covalent); MOPAC and AIMNet2+MOPAC (re-reported MOPAC pairs)	–
MOPAC AO-pair message	Pair scalar, vector and rank-2 reductions; separate AO tensor-product message; reported-pair mask; learned scalar gate from concatenated edge features	MOPAC; AIMNet2+MOPAC	–

Table 4: ShieldingTfnModel forward inputs.

Broad MOPAC input to ShieldingTfnModel

Which block?	What reaches the forward call?	What are the node widths (scalar/flag/vector/ T_2)?	What are the pair widths (radial/scalar/vector/ T_2)?
PM7/MOZYME atomic state	Coulson charge; bond valency; s , p and d populations; Lewis-bond partner count	6 / 0 / 0 / 0	0 / 0 / 0 / 0
Coulson electrostatics	Five field scalars; one aromatic-source count; total, backbone, side-chain and aromatic field vectors; four potential Hessians; two H-parent applicability flags	6 / 2 / 4 / 4	0 / 0 / 0 / 0
Density and Wiberg terms	Five atom-level Wiberg reductions; Wiberg-weighted direction and alignment; atom s - p and traceless p - p reductions. Pairs: Wiberg order, nine squared AO shell-block norms, two scalar, three vector and one rank-2 AO-density reduction	5 / 0 / 2 / 2	10 / 2 / 3 / 1
Non-one-bond peptide carbonyl	Wiberg-weighted carbonyl-response T_0 , T_1 and T_2	1 / 0 / 1 / 1	0 / 0 / 0 / 0
One-bond peptide carbonyl	Wiberg-weighted carbonyl-response T_0 , T_1 and T_2	1 / 0 / 1 / 1	0 / 0 / 0 / 0
Additional source responses	Eight Wiberg-weighted $T_0/T_1/T_2$ source groups; two near-field counts; nearest accepted C=O/C-N distances and their Wiberg-weighted source responses; nearest-C=O direction; two applicability flags; pair Lewis Boolean	14 / 2 / 11 / 10	1 / 0 / 0 / 0
Localised orbitals	Thirteen occupied- and thirteen virtual-LMO scalar reductions	26 / 0 / 0 / 0	0 / 0 / 0 / 0
Global broadcasts	Heat of formation; heat per atom; point-charge dipole; hybridisation dipole; raw-coordinate centroid	2 / 0 / 3 / 0	0 / 0 / 0 / 0
Broad-MOPAC addition	–	61 / 4 / 22 / 18	11 / 2 / 3 / 1
With shared structure	–	96 / 12 / 23 / 18	11 / 2 / 3 / 1

Table 5: Broad MOPAC inputs to the MOPAC and AIMNet2+MOPAC forms of ShieldingTfnModel.

Shared structural inputs

Which block?	What values enter?	In what form?	Where are they produced?
Categorical identity (17)	Element; IUPAC atom; IUPAC residue; AMBER residue; terminal state; residue variant; formal charge; polar-H kind; atom role; hybridisation; backbone role; planar group; locant; ring position; aromatic; exchangeable; IUPAC residue:atom pair	Learned categorical embeddings. The shift readout reuses element, IUPAC atom/pair, atom and backbone roles, AMBER residue, terminal state, variant, formal charge, polar-H kind and exchangeability.	<code>atoms_category_info.npy</code> ; pair formed from IUPAC residue and atom
Positions and graph	Per-atom Cartesian position; covalent bond order and category; same-residue and sequential-residue relations	Positions centred per protein. Spatial edges for $r < 8 \text{ \AA}$, at most 64 neighbours per atom; covalent bonds always included. Ten smooth radial values and real spherical harmonics of orders 0, 1 and 2 are formed at run time.	<code>pos.npy</code> , <code>bonds.npy</code> , <code>residues.npy</code> ; <code>shielding_model.py</code>
Five valence angles	$\angle \text{N-C}\alpha\text{-C}$, $\angle \text{N-C}\alpha\text{-C}\beta$, $\angle \text{C}\beta\text{-C}\alpha\text{-C}$, $\angle \text{C}_{i-1}\text{-N-C}\alpha$ and $\angle \text{C}\alpha\text{-C-N}_{i+1}$	Five scalars in radians, broadcast from the residue to its atoms	<code>LocalBackboneGeometryResult</code>
Six torsions	$\phi, \psi, \chi_1, \chi_2, \chi_3, \chi_4$	$\cos \theta$ and $\sin \theta$: 12 dimensionless scalars; undefined torsions are zero after normalisation	<code>DsspResult</code> ; project IUPAC sign convention
Peptide ω	Signed $\text{C}\alpha_i\text{-C}_i\text{-N}_{i+1}\text{-C}\alpha_{i+1}$ torsion; X-Pro flag	$\cos \omega$, $\sin \omega$ and one binary scalar	<code>PlanarGeometryResult</code>
Pyramidalisation	Absolute displacement of an applicable sp^2 centre from the plane through its three bonded neighbours	One non-negative scalar in $\text{\AA}$; one applicability flag	<code>PlanarGeometryResult</code>
Corrected $C\beta$ residual	$\mathbf{r}_{C\beta} - \mathbf{r}_{C\beta}^{\text{ideal}}$, where $\mathbf{r}_{C\beta}^{\text{ideal}} = \mathbf{r}_{C\alpha} - 0.58273431(\mathbf{c} \times \mathbf{n}) - 0.56802827\mathbf{n} - 0.54067466\mathbf{c}$, $\mathbf{n} = \mathbf{r}_N - \mathbf{r}_{C\alpha}$, $\mathbf{c} = \mathbf{r}_C - \mathbf{r}_{C\alpha}$	One Cartesian vector in $\text{\AA}$; one applicability flag	<code>LocalBackboneGeometryResult</code> ; corrected geometry overlay
DSSP SS8	H, G, I, E, B, T, S and C	Eight one-hot scalars	<code>DsspResult</code> using <code>libdssp</code> [1]
DSSP H bonds and PPII	Two acceptor and two donor H-bond energies; PPII status	Four kcal mol^{-1} scalars; PPII is 1, 0, or -1 when unobserved	<code>DsspResult</code>
Whole-protein formal charge	Sum of per-atom formal-charge labels	One scalar broadcast to all atoms; structural, not a MOPAC value	<code>model_graph.py</code>
Forward applicability (8)	Corrected $C\beta$ residual; six torsions; pyramidalisation	Eight Boolean node inputs in every single-frame graph; retained per frame by <code>ShiftMDModel</code>	<code>graph.py</code>
ShiftMD numeric validity (28)	Twenty-seven structural-scalar validity values; one $C\beta$ -vector validity value	Boolean per-frame inputs to <code>ShiftMDModel</code> only	<code>trajectory_parent.py</code> ; <code>trajectory_inputs.py</code>

Table 6: Shared structural inputs.

ShiftTfnModel input boundaries

Which model?	What direct graph enters?	What other values enter?	What is not a direct input?
ShiftTfnModel (Static)	Typed Graph from one static structure	151 periodic values/supports; nine calculated physical values; one 384-value OpenFold3 residue representation	Transferred shielding-encoder parameters; direct MOPAC or AIMNet2 values; Larsen values; physical-support masks
ShiftTfnModel (First retained frame)	Typed Graph at 150 ps	151 periodic values/supports; nine calculated physical values	Transferred shielding-encoder parameters; OpenFold3; direct MOPAC or AIMNet2 values; Larsen values; physical-support masks

Table 7: Selected ShiftTfnModel forward inputs.

Which input condition?	What are the node widths (scalar/flag/vector/ T_2)?	What are the MOPAC-pair widths (radial/scalar/vector/ T_2)?	What starts the encoder?	What reaches the physical columns?
Typed Graph	35 / 8 / 1 / 0	0 / 0 / 0 / 0	Typed Graph ShieldingTfnModel	Nine calculated values
AIMNet2	291 / 8 / 1 / 0	0 / 0 / 0 / 0	AIMNet2 ShieldingTfnModel	Nine calculated values
MOPAC	53 / 10 / 8 / 7	10 / 2 / 3 / 1	MOPAC ShieldingTfnModel	Nine calculated values
AIMNet2+MOPAC	309 / 10 / 8 / 7	10 / 2 / 3 / 1	AIMNet2+MOPAC ShieldingTfn-Model	Nine calculated values

Table 8: Four-representation ShiftTfnModel inputs. Common inputs: 17 labels, 151 periodic values/supports and, for the Static form, OpenFold3.

Which final Static condition?	What direct electronic input enters?	What starts the encoder?	What reaches the physical columns?
Plain structural/chemical input	–	Fresh parameters	Nine exact zeros
ORCA-shielding transfer	–	Typed Graph ShieldingTfnModel parameters	Nine exact zeros
Direct MOPAC input	Abbreviated MOPAC surface	Fresh parameters	Nine exact zeros
Direct AIMNet2 embedding	256-value AIMNet2 atom representation	Fresh parameters	Nine exact zeros

Table 9: Final four-source ShiftTfnModel (Static) inputs. Common inputs: shared graph, 151 periodic values/supports and OpenFold3.

Abbreviated MOPAC input to ShiftTfnModel

Which block?	What reaches the forward call?	What are the node widths (scalar/flag/vector/ T_2)?	What are the pair widths (radial/scalar/vector/ T_2)?
PM7/MOZYME atomic state	Coulson charge; bond valency; s , p and d populations; Lewis-bond partner count	6 / 0 / 0 / 0	0 / 0 / 0 / 0
Coulson electrostatics	Five field scalars; one aromatic-source count; total, backbone, side-chain and aromatic field vectors; four potential Hessians; two H-parent applicability flags	6 / 2 / 4 / 4	0 / 0 / 0 / 0
Density and Wiberg terms	Five atom-level Wiberg reductions; Wiberg-weighted direction and alignment; atom s - p and traceless p - p reductions. Pairs: Wiberg order, nine squared AO shell-block norms, two scalar, three vector and one rank-2 AO-density reduction	5 / 0 / 2 / 2	10 / 2 / 3 / 1
Non-one-bond peptide carbonyl	Wiberg-weighted carbonyl-response T_0 , T_1 and T_2	1 / 0 / 1 / 1	0 / 0 / 0 / 0
Abbreviated-MOPAC addition	–	18 / 2 / 7 / 7	10 / 2 / 3 / 1
With shared structure	–	53 / 10 / 8 / 7	10 / 2 / 3 / 1

Table 10: Abbreviated MOPAC inputs to ShiftTfnModel. No one-bond carbonyl, additional-response, LMO, global-broadcast or pair Lewis Boolean value enters.

Inputs added for chemical-shift prediction

Which block?	What values enter?	In what form?	Which model receives them?
Periodic side-chain terms	28 values and four support flags	For each χ_k , $k = 1, \dots, 4$: $\cos \chi_k$, $\sin \chi_k$, $\cos 2\chi_k$, $\sin 2\chi_k$, and three soft rotamer weights centred at 180° , 60° and -60° . Weights are a softmax of $4 \cos(\chi_k - \mu_j)$.	ShiftTfnModel (Static), ShiftTfnModel (First retained frame) and ShiftMDModel
Coupled torsions	112 values and seven support flags	All 4×4 products of the first/second circular harmonics for seven pairs: (χ_1, χ_2) , (ϕ, χ_1) , (ψ, χ_1) , (ϕ, χ_2) , (ψ, χ_2) , (χ_2, χ_3) and (χ_3, χ_4) . These are deterministic expansions of six measured torsions, not 112 additional measurements.	ShiftTfnModel (Static), ShiftTfnModel (First retained frame) and ShiftMDModel
Shielding prediction (3)	Predicted isotropic shielding σ_{iso} ; rank-2 magnitude q ; rank-2 shape s	Selected models use the frozen Typed Graph ShieldingTfnModel; the four-representation comparison uses its matching ShieldingTfnModel. Each predicts σ_{iso} and a traceless symmetric tensor T : $q = \sqrt{\text{tr}(T^2)}$ and $s = \sqrt{6} \text{tr}(T^3) / \text{tr}(T^2)^{3/2}$, with $s = 0$ when q is numerically zero.	Selected ShiftTfnModel forms; four-representation comparison; every ShiftMDModel frame; not the final four-source Static comparison
Aromatic ring correction (1)	One calibrated T0 correction in ppm	Candidate-A circular double-loop response from geometry and topology; selected types are Phe benzene, Tyr phenol, Trp benzene, Trp pyrrole and HIE imidazole. Ring atoms and bonded neighbours are excluded. A residue:atom calibration supplies the scalar input.	Selected ShiftTfnModel forms; four-representation comparison; every ShiftMDModel frame; not the final four-source Static comparison
Raw aromatic ring response (1)	Sum of Candidate-A isotropic responses for the same five ring types	Both arms receive 17 labels, coordinates and covalent topology, with no numeric node or pair channels. The control receives exact zero; the treatment fits one intercept-free coefficient per nine-role output. No shielding parent or residual calibration.	Ring-current ShiftTfnModel contrast only
Fixed anisotropy corrections (5)	Peptide C–N, backbone other, side-chain other, backbone X–H and side-chain X–H corrections in ppm	Five separately calibrated T0 responses from the fixed-geometry McConnell calculation. The five values remain separate model inputs.	Selected ShiftTfnModel forms; four-representation comparison; every ShiftMDModel frame; not the final four-source Static comparison
OpenFold residue representation (384)	<code>si_trunk</code> for each residue from OpenFold3 seed 42 [6]	Nucleus-specific LayerNorm $\rightarrow$ Linear(384, 32) $\rightarrow$ SiLU; the 32 values are broadcast to atoms in that residue. <code>zij_trunk</code> is not used.	ShiftTfnModel (Static) only

Table 11: Additional ShiftTfnModel and ShiftMDModel inputs.

Which block is normalised?	What rule is fitted?
Structural graph	Per-channel fitting-member statistics. Invalid numeric values are set to zero after normalisation; applicability is retained separately.
Periodic expansions	Fitted on unique fitting residues, using only valid torsions. Missing features are zero and retain a support flag.
Three shielding values	Centred and scaled separately for H, C and N using source values and element labels; no shift target is used. The final single-conformation fits use 173 proteins; the frozen parent of the retained 100-frame run uses 106.
Six calibrated corrections	Zero-centred and divided by the root-mean-square of nonzero source values from the same fitting population. Exact zeros remain zero.

Table 12: Fitting-only input normalisation.

Which non-input operation follows ShiftTfnModel and ShiftMDModel?	What is the exact operation?
Fitted baseline	Nucleus $\rightarrow$ local chemistry (element, atom role, hybridisation, backbone role, planar group, aromaticity, formal charge, polar-H kind, exchangeability) $\rightarrow$ IUPAC residue:atom $\rightarrow$ prepared state (AMBER residue, terminal state, residue variant); most specific available
Role scale	Raw 1.4826 MAD of fitting residuals for each of nine output roles; no nucleus floor or shrinkage
Reported shift	$\delta_{\text{ppm}} = b(\text{labels}) + s_{\text{role}} y_{\text{norm}}$

Table 13: Output conversion, listed separately from predictor inputs.

Inputs to ShiftMDModel

Which prepared block?	What is its width?	What values enter?	Where are they produced?
Frame axis	100	Frames 15, 30, . . . , 1500 at 10 ps per source frame: 150, 300, . . . , 15,000 ps	H5 trajectory reader
Aligned position	3 per atom	Cartesian coordinates after proper Kabsch SVD alignment. The fitting subset comes from <code>/trajectory/rmsd_tracking/atom_indices</code> ; the iterative mean reference starts at the first retained frame, stops at 10^{-4} Å or 20 iterations, and applies no temporal smoothing.	H5 reader; <code>TransformedConformation</code> and export contract
Fixed identity and topology	11 labels; protein-specific bonds	The 11 direct shift labels listed on the shared-structure page; bond index, order and category. Identity and bonds are fixed across frames.	Static extraction sidecars
Structural scalar	27 values + 27 validity flags	The shared structural scalars except the eight SS8 columns; fixed fitting-set normalisation applied at each frame	Frame extraction
Secondary structure	8	DSSP SS8, kept separate from the 27 scalar values	Frame extraction
Structural vector	3 + one validity flag	Fitting-normalised corrected $C\beta$ residual, rotated into the aligned frame	Frame extraction; trajectory reader
Applicability	8	$C\beta$, six torsions and pyramidalisation flags	Frame extraction
Physical values	9	Three shielding summaries, one ring-current correction and five fixed-anisotropy corrections from the chemical-shift input table	Frozen Typed Graph ShieldingTfnModel (three); calibrated calculations (six)
Periodic values	28 + 4; 112 + 7	Side-chain terms and coupled terms with their support flags	Deterministic per-frame expansion
Torsion history	6 × 2 + 6	Unit cosine/sine pairs for six torsions; six validity flags	Frame extraction
Frozen ShiftTfnModel response	1 + 3 × 36	Normalised parent prediction; 36 invariant hidden-state values for each H/C/N branch	Frozen ShiftTfnModel (First retained frame) evaluated at every frame

Table 14: Prepared ShiftMDModel arrays.

Which per-frame path?	What is its width?	What values enter before pooling?
Response	37	Selected nucleus’s 36 frozen-parent invariants plus the normalised parent prediction
Scalar	62	27 structural scalars, 27 scalar-validity flags and eight applicability flags
Covariant	3	$\log(1 + \ v_t\)$, $\operatorname{asinh}(v_t \cdot v_{150\text{ps}})$ and vector validity
Physical	9	Three shielding and six calibrated correction values
Rotamer	151	28 side-chain terms, four supports, 112 coupled terms and seven supports
Motion	4	Aligned displacement from 150 ps divided by 5 Å, plus $\log(1 + \ \Delta r/(5\text{Å})\ ^2)$
State	54	27 numeric-validity, eight applicability, four rotamer-main supports, seven rotamer-coupling supports and eight SS8 values

Table 15: Seven per-frame paths in ShiftMDModel.

Operations inside ShiftMDModel

Which operation?	What does it do?
Frame contrast	Each path is encoded by a bias-free MLP, then the encoded 150 ps value is subtracted.
Exchangeable summaries	First and second moments over the 100 frame contrasts.
Rotamer ordered summary	Six torsions $\times$ six lags: 36 support-weighted $u_t \cdot u_{t+k} - u_t \cdot u_t$ values; four χ angles $\times$ three anchor-relative occupancies: 12; four χ angles $\times$ six transition rates: 24; total width 72.
Motion ordered summary	Six lagged means of $\log(1 + \ \text{step}\ ^2 / (25 \text{ \AA}^2))$; six lagged means of $\text{asinh}(\Delta r_t \cdot \Delta r_{t+k} / (25 \text{ \AA}^2))$: width 12.
State ordered summary	Five groups; one anchor-delta mean and, at each of six lags, absolute transition and co-presence delta: width 65 before projection.
Ordered lags	1, 2, 4, 8, 16 and 32 retained frames: 150–4,800 ps.
High/low frame pools	<code>softmax(score)</code> and <code>softmax(-score)</code> ; ordinary encoded mean subtracted from each weighted encoded mean.
Neighbour step	One post-summary symmetric, degree-averaged covalent-neighbour message; sigmoid gate from learned bond-order and bond-category embeddings.
Zero-dynamic subtraction	$f(\text{dynamic, typed}) - f(0, \text{typed})$.
Output base	Learned correction added to the uniform mean of 100 normalised frozen-parent predictions.

Table 16: ShiftMDModel forward operations.

Producers and direct method references

Which input?	Where is it produced or read?	Which direct method source applies?
Atom/residue identity, positions and bonds	<code>nmr_extract</code> atom, residue and bond arrays; <code>graph.py</code> ; <code>shielding_model.py</code>	–
Five angles and corrected $C\beta$ vector	<code>LocalBackboneGeometryResult.cpp</code> ; corrected geometry overlay	–
Torsions, SS8 and H bonds	<code>DsspResult.cpp</code> ; <code>libdssp</code>	DSSP [1]
PPII status	<code>DsspResult.cpp</code> ; <code>libdssp</code>	–
Omega and pyramidalisation	<code>PlanarGeometryResult.cpp</code>	–
Equivariant spatial graph	<code>shielding_model.py</code> ; <code>message_passing.py</code> ; <code>e3nn</code>	e3nn [2]
AIMNet2 atom representation	Saved AIMNet2 internal atom representation; model-input assembly	AIMNet2 [3]
MOPAC node and pair values	<code>nmr_extract</code> MOPAC reducers; model-input assembly	MOPAC and PM7 [4, 5]
Shielding summaries	Frozen Typed Graph ShieldingTfnModel for selected/full and ShiftMDModel; matching ShieldingTfnModel for the representation comparison; <code>tensors.py</code>	–
Ring-current correction	<code>ring_current.py</code> ; <code>shielding_residuals.py</code>	–
Fixed anisotropy corrections	Current fixed McConnell producer; <code>shielding_residuals.py</code>	–
OpenFold <code>si_trunk</code>	OpenFold3 seed-42 latent; <code>single_conformation_data.py</code>	OpenFold3 [6]
Aligned 100-frame values	H5 reader export; <code>TransformedConformation</code> ; <code>trajectory_parent.py</code> ; <code>trajectory_model.py</code>	–

Table 17: Input producers and direct method sources. A dash marks no external literature assignment.

References

- [1] Kabsch and Sander (1983), *Biopolymers* 22, 2577–2637. doi:10.1002/bip.360221211.
- [2] Geiger and Smidt (2022), *e3nn: Euclidean Neural Networks*, arXiv:2207.09453.
- [3] Anstine, Zubatyuk and Isayev (2025), *Chem. Sci.* 16, 10228–10244. doi:10.1039/D4SC08572H.
- [4] Moussa and Stewart (2026), *J. Open Source Softw.* 11, 8025. doi:10.21105/joss.08025.
- [5] Stewart (2013), *J. Mol. Model.* 19, 1–32. doi:10.1007/s00894-012-1667-x.
- [6] OpenFold3 Team (2026), *OpenFold3 Preview 2 Technical Report*. doi:10.5281/zenodo.19001000.